Autonomous Racing Navigation:
Experimental Results Summary

AutoCar Project

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Abstract

This document consolidates lap-time and tracking metrics from all navigation experiments conducted on the AutoCar simulator. Experiments progress in three stages: (i) algorithm selection on the *circle circuit* (Stanley, MPC, Pure Pursuit), (ii) Pure Pursuit tuning on the *F1 circuit* (Albert Park), and (iii) the same F1 circuit with the Pure Pursuit + LiDAR navigation stack, including robustness sweeps over perception latency and odometry noise. All raw data are archived under `results/`.

Contents

1 Master Comparison Table

Table 4 consolidates **all** recorded runs in a single view. Lap times from manual sessions report T_{min} (best lap) with mean ± std in parentheses; benchmark sessions report post-warmup median T_{med}. Best result per stage is marked **bold**. Rows marked TIMEOUT did not complete a measured lap within the allotted time.

Table 1: Stage I — Algorithm selection on the circle circuit (2026-05-25).

#	Algorithm	Line	T_{min} (s)	$\bar{T} \pm \sigma$ (s)	$\bar{v}$ (m/s)
1	Stanley	centerline	190.1	191.1±0.9	3.42
2	Stanley	racing	172.6	173.1±0.7	3.63
3	MPC	centerline	108.9	111.1±2.2	5.89
4	MPC	racing	108.5	110.4±1.8	5.71
5	Pure Pursuit	centerline	99.9	105.4±4.8	6.23
6	Pure Pursuit	racing	**92.5**	95.0±2.2	6.63

Table 2: Stage II — Pure Pursuit on the F1 circuit (2026-06-08).

#	Line	Profile	L (ms)	σ (m)	T_{med} (s)	$\bar{v}$ (m/s)	$\varepsilon_{\mathrm{rms}}$ (m)	$\dot{\delta}_{\mathrm{max}}$ (rad/s)
7	centerline	`baseline`	0	0	174.6	3.52	0.768	12.46
8	racing	`racing`	0	0	89.5	6.41	0.295	5.07
9	racing	`finetuned (conservative)`	0	0	**86.7**	6.61	0.351	**1.33**
10	racing	`finetuned (aggressive)`	0	0	TIMEOUT	–	–	–
11	racing	`finetuned_perturbed`	200	0.1	98.7	18.55†	0.619	3.00

Conservative (`f1_finetuned`): `cruise_velocity` = 8.0 m/s, `max_lateral_accel` = 5.5 m/s^2. *Aggressive* (tuning rejects, `docs/rapport_optimisation_f1.md`): `cruise_velocity` 9–10 m/s and/or `max_lateral_accel` = 6.5 m/s^2 → crash or off-track, no lap-time gain; PP tracking ceilings at ∼8 m/s.

Table 3: Stage III — Pure Pursuit + LiDAR on the F1 circuit (2026-06-10).

#	Profile	L (ms)	σ (m)	T_{med} (s)	$\bar{v}$ (m/s)	$\varepsilon_{\mathrm{rms}}$ (m)	$\dot{\delta}_{\mathrm{max}}$ (rad/s)	Off
12	`baseline`	0	0	125.6	4.17	0.071	0.80	0
13	`finetuned`	0	0	**115.1**	4.71	0.108	**0.60**	0
14	`l0_n005`	0	0.05	TIMEOUT	–	–	–	–
15	`l0_n01`	0	0.10	139.6	6.00	0.155	0.60	0
16	`l200_n0`	200	0	TIMEOUT	–	–	–	–
17	`l200_n005`	200	0.05	250.9	3.32	0.367	0.60	0
18	`l200_n01`	200	0.10	501.0	3.84	0.143	0.60	0
19	`l500_n0`	500	0	TIMEOUT	–	–	–	–
20	`l500_n005`	500	0.05	TIMEOUT	–	–	–	–
21	`l500_n01`	500	0.10	TIMEOUT	–	–	–	–

Source: `results/benchmark_2026-06-10T10-52-32/summary.csv`; $\dot{\delta}_{\mathrm{max}}$ from post-warmup lap 2 (`steering_rate_max` in session `lap_times.csv`).

Table 4: Master comparison — best lap time per stage (Tables 1–3).

Stage	Track	Stack	Best lap (s)
I	Circle circuit	Stanley / MPC / Pure Pursuit	**92.5**
II	F1 circuit	Pure Pursuit	**86.7**
III	F1 circuit	Pure Pursuit + LiDAR	**115.1**

†Stage II perturbed run: inflated path distance (1831 m vs. ≈573 m) under localisation noise; $\bar{v}$ is not comparable to nominal runs. Stage I lap column format: $T_{\min}$ ($\bar{T} \pm \sigma$). Stages II–III: T_{med} and $\dot{\delta}_{\max}$ from post-warmup lap 2 (warmup lap 1 excluded from summary).

2 Experimental Setup

2.1 Tracks and navigation stacks

- **Circle circuit** — oval track used for controller comparison (`results/stanley_2026-05-25T13-55-37`, etc.). Nominal lap length from waypoint polylines (`waypoints.csv` / `waypoints_racing.csv`): centerline ≈651 m, racing line ≈627 m.
- **F1 circuit** — Albert Park layout (scaled for simulation), used in both Stage II and Stage III (`results/benchmark_2026-06-08` `results/benchmark_2026-06-10T10-52-32`). Nominal lap length from precomputed waypoint polylines (`waypoints_f1.csv` / `waypoints_f1_racing.csv`; Stage II only): centerline ≈596 m, racing line ≈568 m (real Melbourne GP circuit: ≈5.3 km). Stage II employs the standard Pure Pursuit stack with these CSV paths loaded at launch. Stage III employs Pure Pursuit + LiDAR: lap 1 explores via LiDAR; centerline and racing line are *computed online* from the SLAM map and stored in memory (not read from CSV). The benchmark config label `f1_circuit_fenced` refers to this LiDAR-enabled setup on the same circuit geometry, not a separate track.

2.2 Trajectory modes

- **Stages I–II (open-loop stacks)** — `centerline` and `racing` refer to precomputed waypoint files (`waypoints*.csv`) loaded at launch.
- **Stage III (Pure Pursuit + LiDAR)** — the `line` launch flag selects the same nominal mode names, but trajectories are *computed at runtime* and held in memory: lap 1 builds a local centerline from LiDAR (Follow-the-Gap) and records the exploration path into the SLAM map; at lap 2 the `global_planner_lidar` node derives a closed-loop centerline from that path, optimises a min-curvature racing line, smooths it, and caches the result for subsequent laps (typical lap-2+ distance ≈520–540 m in our runs, vs. ≈568 m for the offline `waypoints_f1_racing.csv`).

2.3 Metrics

- **Lap time** T_{lap} (s) — elapsed time between consecutive start/finish crossings.
- **Average speed** $\bar{v}$ (m/s) and **peak speed** $v_{\max}$ (m/s).
- **Path distance** d (m) — integrated odometry over the lap.
- **Lateral error RMS** ε_{rms} (m) and **peak lateral error** $\varepsilon_{\max}$ (m) — benchmark runs only.
- **Peak steering rate** $\dot{\delta}_{\max}$ (rad/s) — maximum $|\mathrm{d}\delta/\mathrm{d}t|$ per lap (proxy for steering oscillation).
- **Off-track events** — count of boundary violations (benchmark runs only).

2.4 Recording conventions

Manual launches write per-lap records to `results/<stack>_<run_id>/lap_times.csv` (15 columns when extended metrics are enabled). Batch benchmarks (`scripts/benchmark.py`) aggregate results into `summary.csv` with one warmup lap excluded from reported statistics (`warmup_laps = 1`). Runs that did not complete a measured lap within the allotted time are marked **timeout**.

3 Algorithm Selection — Circle Circuit

Three path-tracking stacks were evaluated on the circle circuit under both centerline and racing trajectories. Sessions were recorded on 2026-05-25. Pure Pursuit achieved the lowest lap times and was retained for subsequent experiments.

3.1 Summary comparison

Table 5 reports aggregate statistics over all recorded laps. Racing-line trajectories shorten the path (≈630 m vs. ≈653 m) and generally reduce lap time.

Table 5: Circle-circuit algorithm comparison (2026-05-25). Best lap time per trajectory mode in **bold**.

Stack	Line	n	$\bar{T}_{\text{lap}}$	σ_T	$T_{\min}$ (s)	$T_{\max}$	$\bar{v}$	$v_{\max}$ (m/s)
Stanley	centerline	3	191.1	0.9	190.1	191.8	3.42	5.85
Stanley	racing	2	173.1	0.7	172.6	173.6	3.63	5.91
MPC	centerline	3	111.1	2.2	**108.9**	113.2	5.89	7.58
MPC	racing	5	110.4	1.8	**108.5**	113.0	5.71	7.56
Pure Pursuit	centerline	3	105.4	4.8	99.9	109.0	6.23	7.60
Pure Pursuit	racing	3	**95.0**	2.2	**92.5**	96.8	6.63	7.67

3.2 Racing-line speedup over centerline

Table 6: Racing-line improvement relative to centerline on the circle circuit.

Stack	$\bar{T}_{\text{center}}$ (s)	$\bar{T}_{\text{racing}}$ (s)	ΔT (%)
Stanley	191.1	173.1	−9.4
MPC	111.1	110.4	−0.6
Pure Pursuit	105.4	95.0	**−9.8**

3.3 Per-lap records

3.3.1 Stanley

Table 7: Stanley — circle circuit. Source: `results/stanley_2026-05-25T13-55-37` (centerline), `results/stanley_racing_2026-05-25T13-48-10` (racing).

Line	Lap	T_{lap} (s)	$\bar{v}$ (m/s)	$v_{\max}$ (m/s)	d (m)
centerline	1	191.3	3.41	5.83	652.5
centerline	2	190.1	3.43	5.85	652.5
centerline	3	191.8	3.40	5.84	652.8
racing	1	172.6	3.64	5.83	628.3
racing	2	173.6	3.62	5.91	628.6

3.3.2 MPC

Table 8: MPC — circle circuit. Source: `results/mpc_2026-05-25T16-46-36` (centerline), `results/mpc_racing_2026-05-25T16-56-47` (racing).

Line	Lap	T_{lap} (s)	$\bar{v}$ (m/s)	$v_{\max}$ (m/s)	d (m)
centerline	1	113.2	5.78	7.46	653.9
centerline	2	111.1	5.89	7.58	654.0
centerline	3	108.9	6.00	7.41	653.6
racing	1	113.0	5.58	7.31	630.6
racing	2	108.5	5.80	7.51	629.7
racing	3	110.4	5.71	7.45	630.2
racing	4	110.8	5.69	7.49	630.3
racing	5	109.1	5.78	7.56	630.1

3.3.3 Pure Pursuit (selected algorithm)

Table 9: Pure Pursuit — circle circuit. Source: `results/pure_pursuit_2026-05-25T16-28-02` (centerline), `results/pure_pursuit_racing_2026-05-25T16-34-12` (racing).

Line	Lap	T_{lap} (s)	$\bar{v}$ (m/s)	$v_{\max}$ (m/s)	d (m)
centerline	1	99.9	6.55	7.60	654.1
centerline	2	107.2	6.12	7.56	656.4
centerline	3	109.0	6.03	7.58	657.2
racing	1	96.8	6.51	7.52	629.7
racing	2	95.7	6.58	7.50	629.5
racing	3	92.5	6.80	7.67	629.0

3.4 Selection outcome

Pure Pursuit delivered the **fastest best lap** (92.5 s, racing line) and the **largest racing-line gain** (−9.8%). Stanley was approximately 80 s slower per lap; MPC was competitive on centerline but did not outperform Pure Pursuit. Pure Pursuit was therefore chosen for all F1-circuit experiments.

4 F1 Circuit — Pure Pursuit Progression

Four configurations were benchmarked on the Albert Park F1 circuit on 2026-06-08 (`results/benchmark_2026-06-08T19-47-00`). Each run completed two laps with one warmup lap excluded from the summary.

Table 10: Pure Pursuit on the F1 circuit: baseline, racing line, parameter finetuning, and finetuned stack under injected latency/noise. Source: `results/benchmark_2026-06-08T19-47-00/summary.csv`.

Profile	Line	Lat. (ms)	Odom. σ	T_{med} (s)	$\bar{v}$ (m/s)	$v_{\max}$ (m/s)	ε_{rms} (m)	$\dot{\delta}_{\max}$ (rad/s)
baseline	centerline	0	0.0	174.6	3.52	7.54	0.768	12.46
racing	racing	0	0.0	89.5	6.41	7.63	0.295	5.07
finetuned	racing	0	0.0	**86.7**	6.61	7.69	0.351	**1.33**
finetuned_perturbed	racing	200	0.1	98.7	18.55†	7.32	0.619	3.00

†Elevated $\bar{v}$ reflects inflated path distance (1831 m vs. ≈573 m) under perturbation, not higher on-track speed.

4.1 Per-lap records

Table 11 lists both laps for each Stage II run (warmup lap 1, measured lap 2). Sources: `results/pure_pursuit_2026-06-08T19-47-0` (baseline), `results/pure_pursuit_racing_2026-06-08T19-53-41` (racing), `results/pure_pursuit_racing_2026-06-08T19-57-17` (finetuned), `results/pure_pursuit_racing_2026-06-08T20-00-46` (perturbed).

Table 11: F1 circuit — per-lap records (2026-06-08).

Profile	Lap	T_{lap} (s)	$\bar{v}$ (m/s)	$v_{\max}$ (m/s)	$\varepsilon_{\mathrm{rms}}$ (m)	$\dot{\delta}_{\max}$ (rad/s)	Off
baseline	1	169.2	3.61	7.65	0.711	7.71	0
baseline	2	174.6	3.52	7.54	0.768	12.46	0
racing	1	95.6	6.04	7.66	0.561	14.79	0
racing	2	89.5	6.41	7.63	0.295	5.07	0
finetuned	1	87.8	6.52	7.74	0.359	0.80	0
finetuned	2	**86.7**	6.61	7.69	0.351	**1.33**	0
perturbed	1	99.9	18.49	7.32	0.677	3.71	0
perturbed	2	98.7	18.55	7.32	0.619	3.00	0

4.2 Finetuned profile — parameter changes (Stage II)

Navigation overrides are archived under `results/benchmark_2026-06-08T19-47-00/nav_overrides/` (materialised from `scripts/configs/f1_circuit.yaml`). The open-loop Pure Pursuit stack uses precomputed waypoints; the `finetuned` profile (`f1_finetuned`) builds on `racing` (`f1_racing`) with one additional path-tracker change (Table 12). Stage III finetuned parameters (LiDAR stack) are documented separately in Section 5.2 — the same profile name, but a different tuning set.

Table 12: Stage II parameter deltas for the F1 finetuned profile (canonical: `f1_circuit.yaml`; archived as `run_001` `f1_baseline`, `run_002` `f1_racing`, `run_003` `f1_finetuned`).

Parameter	Baseline	Racing	Finetuned	Role
Trajectory (`line`)	centerline	racing	racing	Switch to the min-curvature QP racing line (`waypoints_f1_racing.csv`), smoothed offline (`smooth_racing_line.py`).
`curvature_lookahead`	140	60	60	Shorter bend-detection horizon on the twisty F1 layout: brake later instead of treating the whole circuit as one continuous corner.
`heading_soft`	0.6	1.2	1.2	Relax heading-error penalty so throttle is not crushed in corners (baseline setting cut speed to ≈16% of target).
`lookahead_curv_extra`	0	0	3.0 m	**Finetuned-only:** add extra lookahead on straights only; suppresses high-speed zigzag without widening lookahead in corners.
`lookahead_gain` / `lookahead_min`	0.4 / 1.5	0.3 / 1.0	0.3 / 1.0	Stack default Pure Pursuit gains shortened for F1 tight corners (L_d at 8 m/s: 4.7 m → 3.4 m).

Baseline column: stack defaults (`navigation_params.yaml`). The `f1_baseline` run also uses 0.3 / 1.0 in `f1_circuit.yaml` (F1 circuit override applied before racing-line / finetuning steps).

Incremental change (`racing` → `finetuned`). Only `lookahead_curv_extra` differs between `run_002` and `run_003`; it accounts for the 3.1% lap-time gain (89.5 s → 86.7 s) and the drop in $\dot{\delta}_{\max}$ (5.07 → 1.33 rad/s). All other navigation parameters are identical between the two profiles. `f1_finetuned_perturbed` reuses the `f1_finetuned` navigation block; only `latency_ms` = 200 and `odom_noise_std` = 0.1 differ.

4.3 Key observations

1. Switching from centerline to racing line on F1 cuts median lap time by 49% (174.6 s → 89.5 s).
2. Parameter finetuning yields a further 3.1% reduction (89.5 s → **86.7 s**).
3. Under 200 ms latency and odometry noise ($\sigma = 0.1$ m), the finetuned controller remains stable (zero off-track events) but lap time increases by 13.8% and path distance diverges, indicating degraded localisation.

4. Lateral tracking improves markedly with the racing line: ε_{rms} drops from 0.768 m (centerline) to 0.295 m (racing).
5. Parameter finetuning also smooths steering: peak steering rate on the measured lap falls from 5.07 rad/s (racing) to **1.33 rad/s** (finetuned), a −74% reduction in steering-rate spikes.

5 F1 Circuit — Pure Pursuit with LiDAR Stack

Stage III reuses the same F1 circuit as Stage II, but switches to the Pure Pursuit + LiDAR navigation stack, which constrains the vehicle to LiDAR-detected track boundaries. Ten configurations were evaluated on 2026-06-10 (`results/benchmark_2026-06-10T10-52-32`), sweeping perception latency $L \in \{0, 200, 500\}$ ms and odometry noise $\sigma \in \{0, 0.05, 0.1\}$ m.

5.1 Nominal conditions ($L = 0$, $\sigma = 0$)

Table 13: Pure Pursuit + LiDAR under nominal sensing on the F1 circuit (racing line).

Profile	T_{med} (s)	$\bar{v}$ (m/s)	$v_{\max}$ (m/s)	d (m)	ε_{rms} (m)	$\dot{\delta}_{\max}$ (rad/s)	Off-track
baseline	125.6	4.17	5.77	523.4	0.071	0.80	0
finetuned	**115.1**	4.71	7.25	542.1	0.108	**0.60**	0

Finetuning reduces lap time by 8.4% while maintaining sub-0.11 m lateral error RMS and lowering peak steering rate from 0.80 to **0.60 rad/s** on lap 2 (−25%). Parameter changes vs. baseline are listed in Table 14.

5.2 Finetuned profile — parameter changes (Stage III)

Stage III uses the Pure Pursuit + LiDAR stack: lap 1 builds a SLAM map and extracts a smoothed racing line; lap 2+ races on that online trajectory. Overrides are archived under `results/benchmark_2026-06-10T10-52-32/nav_overrides/` (`run_000` baseline, `run_001` finetuned; source config `f1_pure_pursuit_lidar.yaml`). Unlike Stage II, finetuning here retunes *all three* navigation nodes (local planner, LiDAR global planner, path tracker).

Table 14: Stage III parameter deltas for the F1 finetuned profile (Pure Pursuit + LiDAR; baseline vs. finetuned).

Parameter	Baseline	Finetuned	Role
local_planner			
`cruise_velocity` / `avoid_velocity`	6.0	**7.5 m/s**	Raise lap-2+ target speed; baseline was conservative for SLAM exploration.
`max_lateral_accel`	4.5	5.0	Allow slightly higher cornering speed; 5.5 was too nervous on this stack.
`curvature_lookahead`	140	**110**	Shorten bend horizon (Stage II used 60, but here 60 caused lap-2 speed bobbing).
`curvature_smooth_window`	5	7	Smooth cruise-velocity transitions when curvature estimate changes.
`decel_rate`	6.0	6.5	Slightly faster braking into corners at higher cruise speed.
global_planner_lidar			
`waypoints_ahead` / `waypoints_behind`	5 / 2	6 / 1	More lookahead spline goals, fewer behind — smoother path at lap seam.
`waypoint_search_ahead`	30	35	Wider index search for stable goal selection at 7.5 m/s.
`centerline_step`	3.5	**3.0 m**	Denser lap-1 exploration path → better online racing line.
`racing_mincurv_max_offset`	5.0	4.5 m	Less aggressive corner cutting on the min-curvature racing line.
`racing_smooth_iters` / `racing_smooth_pre_iters`	10 / 4	12 / 5	Extra smoothing of the SLAM-derived racing trajectory.
`racing_smooth_alpha` / `racing_smooth_pre_alpha`	0.40	0.42	Slightly stronger smoothing blend on extracted waypoints.
path_tracker			
`lookahead_gain` / `lookahead_min` / `lookahead_max`	0.4 / 1.5 / 6.0	**0.48 / 1.8 / 8.0**	Longer adaptive lookahead at 7.5 m/s ($L_d \approx 5.4$ m) to cut zigzag.
`lookahead_curv_extra`	0	**4.0 m**	Straight-only lookahead extension (Stage II: 3.0 m; higher speed needs more).
`lookahead_curv_soft`	0.08	0.10	Ignore high-frequency spline-curvature noise when adapting lookahead.
`closest_search_ahead`	120	140	Stabilise closest-point index at higher lap speed.
`steering_limits`	0.95	0.92	Lower steering cap to suppress peak steering commands.
`heading_soft`	0.6	0.65	Mild heading-penalty relax (Stage II used 1.2; here throttle modulation did not fix zigzag).

Perturbed profiles (`finetuned_perturbed_*`). The eight sweep runs share the same navigation block as `finetuned`; only `latency_ms` and `odom_noise_std` differ ($L \in \{0, 200, 500\}$ ms, $\sigma \in \{0, 0.05, 0.1\}$ m).
Unchanged on purpose (both stages). `steer_smoothing` stays at 1.0 (off): low-pass filtering amplifies oscillation via phase lag. Stage III keeps `exploration_velocity` at 3.0 m/s for a stable lap-1 SLAM pass.

5.3 Robustness sweep

Table 15: Pure Pursuit + LiDAR robustness matrix. TIMEOUT = run timed out before completing a measured lap. Source: `results/benchmark_2026-06-10T10-52-32/summary.csv`.

Profile	L (ms)	σ (m)	Status	T_{med} (s)	$\bar{v}$ (m/s)	ε_{rms} (m)	$\dot{\delta}_{\max}$ (rad/s)	Off-track
finetuned_perturbed_l0_n005	0	0.05	TIMEOUT	–	–	–	–	–
finetuned_perturbed_l0_n01	0	0.10	OK	139.6	6.00	0.155	0.60	0
finetuned_perturbed_l200_n0	200	0.00	TIMEOUT	–	–	–	–	–
finetuned_perturbed_l200_n005	200	0.05	OK	250.9	3.32	0.367	0.60	0
finetuned_perturbed_l200_n01	200	0.10	OK	501.0	3.84	0.143	0.60	0
finetuned_perturbed_l500_n0	500	0.00	TIMEOUT	–	–	–	–	–
finetuned_perturbed_l500_n005	500	0.05	TIMEOUT	–	–	–	–	–
finetuned_perturbed_l500_n01	500	0.10	TIMEOUT	–	–	–	–	–

5.4 Robustness summary

- **Latency sensitivity:** all 500 ms configurations timed out. At 200 ms, only runs with $\sigma > 0$ completed; pure-latency injection (L=200 ms, σ=0) failed.
- **Noise-only perturbation** (L=0): moderate noise (σ=0.05 m) caused timeout; σ=0.10 m completed with T_{med} = 139.6 s (+21% vs. finetuned nominal).
- **Combined perturbation** (L=200 ms): lap times scale severely (250.9 s at σ=0.05 m; 501.0 s at σ=0.10 m), yet no off-track events were recorded.
- **Best LiDAR result:** finetuned profile, **115.1 s** median lap under nominal sensing.

6 Cross-Experiment Comparison

Table 4 (Section 1) summarises the best lap time per stage; Tables 1–3 list all individual runs. Direct comparison across stages is indicative only — the circle and F1 circuits differ in layout; Stages II and III share the same F1 circuit but use different navigation stacks.

Table 16: Best lap times by experimental stage (same as Table 4).

Stage	Track	Stack	Best T_{lap} (s)	Source
Algorithm selection	circle	Pure Pursuit	92.5	`results/pure_pursuit_racing_2`
F1 progression	F1 circuit	Pure Pursuit	86.7	`results/pure_pursuit_racing_2`
F1 + LiDAR	F1 circuit	Pure Pursuit + LiDAR	115.1	`results/benchmark_2026-06-10T`

7 Conclusions

1. **Algorithm:** Pure Pursuit outperformed Stanley and MPC on the circle circuit, achieving a best lap of 92.5 s on the racing line.
2. **Racing line:** switching from centerline to racing consistently reduces lap time (−9.8% on circle; −49% on F1).
3. **F1 finetuning:** parameter tuning on Pure Pursuit yields **86.7 s** on the F1 circuit (Stage II) — the overall best result — and reduces peak steering rate from 5.07 to 1.33 rad/s (−74%).
4. **LiDAR stack:** switching from Pure Pursuit to Pure Pursuit + LiDAR on the same F1 circuit increases lap time by ≈28 s (115.1 s vs. 86.7 s, both finetuned, racing line), but peak steering rate stays low (0.60 rad/s vs. 1.33 rad/s in Stage II).
5. **Robustness:** the LiDAR stack tolerates moderate combined perturbation without off-track events, but 500 ms latency is beyond the operational envelope; 200 ms combined with noise degrades performance substantially.

A Data Index

Table 17: Result directories referenced in this report.

Experiment	Directory	Date
Stanley, circle, centerline	`results/stanley_2026-05-25T13-55-37`	2026-05-25
Stanley, circle, racing	`results/stanley_racing_2026-05-25T13-48-10`	2026-05-25
MPC, circle, centerline	`results/mpc_2026-05-25T16-46-36`	2026-05-25
MPC, circle, racing	`results/mpc_racing_2026-05-25T16-56-47`	2026-05-25
Pure Pursuit, circle, centerline	`results/pure_pursuit_2026-05-25T16-28-02`	2026-05-25
Pure Pursuit, circle, racing	`results/pure_pursuit_racing_2026-05-25T16-34-12`	2026-05-25
PP, F1 baseline (centerline)	`results/pure_pursuit_2026-06-08T19-47-06`	2026-06-08
PP, F1 racing line	`results/pure_pursuit_racing_2026-06-08T19-53-41`	2026-06-08
PP, F1 finetuned	`results/pure_pursuit_racing_2026-06-08T19-57-17`	2026-06-08
PP, F1 finetuned + perturbation	`results/pure_pursuit_racing_2026-06-08T20-00-46`	2026-06-08
PP, F1 benchmark (aggregate)	`results/benchmark_2026-06-08T19-47-00`	2026-06-08
Pure Pursuit + LiDAR, F1 circuit	`results/benchmark_2026-06-10T10-52-32`	2026-06-10
PP + LiDAR session runs (×10)	`results/benchmark_2026-06-10T10-52-32/summary.csv` (`run_dir`)	2026-06-10